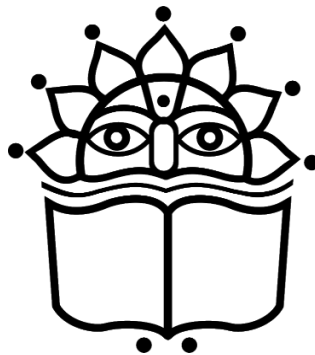


A
PROJECT REPORT
ON
**Design & Development
Of
Support Cleaning Machine
For 3D Printer**

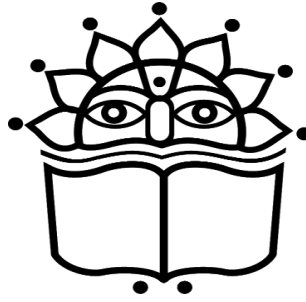
BY
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Of
FINAL YEAR MECHANICAL ENGINEERING

Under The Guidance Of
Prof. S. V. Shelge



Department of Mechanical Engineering
**Vidya Pratishthan's
Kamalnayan Bajaj Institute of Engineering and
Technology, Baramati.
2025-2026**



**Vidya Pratishthan's
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Baramati - 413133**

CERTIFICATE

This is to certify that project titled

“Design & Development Of Support Cleaning Machine For 3D Printer ”

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in fulfillment for the award of Bachelor of Mechanical Engineering of Savitribai Phule Pune University (S.P.P.U) at Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati, is a record of original research work carried out under my supervision.

To the best of our knowledge, the matter embodied in the thesis has not been submitted to any University/Institute for the award of any Degree.

Date: 27/06/2026

Place: Baramati

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ACKNOWLEDGEMENT

We take this opportunity to express our profound gratitude and deepest regards to our guide **Prof. S. V. Shelge** for his exemplary guidance, monitoring, and constant encouragement throughout the course of this project. I owe my gratitude to **Dr. Manisha S. Lande**, Head of Mechanical Engineering Department and Principal, **Dr. S. B. Lande** for providing necessary support in the form of technical facilities in the department and institute. We are also thankful to other professors for their comments and reviews during the progress seminars and non-teaching staff for their help during experimentation.

Last but not the least, we would like to thank our family members and friends for their constant support, advice, and co-operation.

Mr. Ajinkya S. Chavan
Mr. Raviraj V. Kumbhar
Mr. Swapnil B. Yadav

ABSTRACT

The project “Design and Development of Support Cleaning Machine for 3D Printer” aims to create an efficient, low-cost post-processing system for removing support structures from 3D-printed parts. In additive manufacturing, supports are essential but their manual removal is tedious, inconsistent, and may damage delicate components. The proposed machine automates this process using a chemical-based cleaning system integrated with temperature control and agitation for effective support removal. The design procedure included concept generation, component selection, CAD modeling in SolidWorks, gap identification, and feasibility analysis. The optimized system ensures safe, uniform, and damage-free cleaning. Overall, the machine provides a compact, user-friendly, and sustainable solution that enhances post-processing efficiency and supports the advancement of additive manufacturing technology.

Keywords: 3D Printing, Support Removal, Support Cleaning Machine, Chemical Cleaning System, Automated Cleaning, Temperature Control.

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Chapter 1

Introduction

The Additive manufacturing, or 3D printing, enables the creation of complex and precise components but requires support structures that must be removed after printing. Manual removal of these supports is time-consuming, inconsistent, and may damage delicate parts. To overcome this, a Support Cleaning Machine is designed to automate the post-processing stage using chemical cleaning, temperature control, and agitation. The system efficiently dissolves soluble supports like PVA and QSR-30 without affecting part quality. Compact, affordable, and user-friendly, it reduces manual effort and improves surface finish. This innovative machine integrates mechanical, electrical, and thermal systems to enhance efficiency and reliability in 3D printing workflows.



Fig. No.1.1- Support Cleaning Machine for 3D Printer

1.1 Need for Automated Post-Processing in 3D Printing-

In Additive manufacturing, commonly known as 3D printing, has revolutionized the production of complex and customized components across industries such as automotive, medical, aerospace, and consumer goods. However, the post-processing stage, particularly the removal of support material, remains one of the most time-consuming and labor-intensive tasks. During printing, support structures are essential to stabilize overhangs and complex geometries, but their manual removal often damages the part surface and compromises accuracy. Current manual methods using tools or solvents require significant time and operator skill. Hence, an automated system that can safely, efficiently, and uniformly remove support material is necessary. The development of a Support Cleaning Machine ensures consistency, reduces labor dependency, and enhances the surface quality of printed parts, such a machine bridges the gap between low-cost manual techniques and high-end

industrial cleaning systems, enabling faster production cycles and improved part reliability in laboratory.

1.2 Construction Details of Support Cleaning Machine

The support cleaning machine consists of mechanical, thermal, and control subsystems integrated to perform automated cleaning of 3D-printed parts. The primary component is a stainless steel cleaning tank, which holds the cleaning solution and the printed part. A mild steel frame provides structural rigidity and support to the assembly. The heating element, mounted below the tank, maintains the solution temperature within the range of 50°C–70°C, enhancing the dissolution rate of the support material. A digital temperature sensor connected to an Arduino ESP8266-based control unit regulates temperature automatically and provides safety against overheating. The agitation mechanism, driven by a small DC motor, ensures uniform mixing of the cleaning medium to prevent localized saturation. The system uses QSR-30 (EcoWorks Cleaning Agent), a biodegradable and water-soluble chemical effective for dissolving PVA-based supports. The overall construction emphasizes compactness, low cost, easy maintenance, and operational safety for use in laboratories and educational institutes.

1.3 Definition of Automated Support Cleaning Machine (ASCM):

The Automated Support Cleaning Machine (ASCM) is a self-contained system developed to automatically remove support materials from 3D-printed parts through mechanical, thermal, or chemical actions. It is designed to minimize human intervention while maintaining high-quality surface finishing and dimensional precision. The system incorporates a part holding mechanism that securely positions the printed components during the cleaning process.



Fig. No.1.3.1- Automated Support Cleaning Machine (ASCM)

1.4 Role Of The ASCM in Modern 3D Printing

In modern 3D printing production, the Automated Support Cleaning Machine (ASCM) plays a crucial role in enhancing the overall efficiency, precision, and quality of manufactured components. As 3D printing technologies such as FDM, SLA, and SLS evolve to produce complex geometries, the need for effective and reliable post-processing has become increasingly important. The ASCM automates the support removal process, which was traditionally performed manually, saving significant time and labor. By maintaining controlled temperature, agitation, and solvent flow, it ensures uniform cleaning and prevents surface damage or deformation of delicate parts. This results in improved dimensional accuracy and superior surface finish, which are essential for functional prototypes and end-use products. The ASCM contributes to faster production cycles by minimizing downtime between printing and finishing stages. Its automated operation allows multiple parts to be cleaned simultaneously, thereby increasing throughput in industrial and research environments. Moreover, the system enhances safety by reducing direct human contact with heated or chemical solutions. Its programmable control system ensures consistency and repeatability, leading to standardized post-processing results across batches.

1.5 Technological Advancements in the ASCM

In Technological advancements in the Automated Support Cleaning Machine (ASCM) have significantly improved its performance, efficiency, and adaptability in modern additive manufacturing environments. Recent developments integrate smart control systems powered by microcontrollers and PLCs, allowing precise management of temperature, agitation speed, and cleaning duration. Sensor-based monitoring has enhanced operational safety and process consistency by automatically detecting water levels, temperature fluctuations, and mechanical faults. The incorporation of PID temperature control ensures stable thermal conditions, optimizing the dissolution rate of support materials without damaging printed parts. Modern ASCMs now feature variable-speed agitation systems driven by DC or stepper motors, providing adjustable turbulence suited to different material types. The use of ultrasonic transducers in advanced models has further improved cleaning efficiency by generating microscopic cavitation bubbles for deep and uniform cleaning. Touchscreen user interfaces and digital displays make operation intuitive, enabling users to set and monitor all parameters with ease.

Chapter 2

Literature Review

This chapter discusses the previous research paper and their research on the Support Cleaning Machine for 3D Printer. The information obtained from these papers is very helpful for finalizing and specializing the project work and terminologies used for project. The source of the review is collected or studied from journals, websites, and books.

2.1 Literature Review-Need for Support Structures in 3D Printing

The In Additive Manufacturing (AM), support structures are essential to maintain the stability of overhanging surfaces and prevent deformation during the printing process . Supports help control thermal stresses and provide mechanical stability, especially in metal AM processes such as Selective Laser Melting (SLM) and Electron Beam Melting (EBM). However, these structures are non-functional and must be removed after printing, making post-processing a crucial but time-consuming stage.

2.2 Challenges in Support Removal

Manual removal is the most common method, often involving tools like pliers or saws. This process is labor-intensive, may damage parts, and leads to poor surface quality due to residual marks and roughness. For complex geometries, manual removal becomes impractical and inefficient. Thus, researchers have focused on automated and chemical-based removal methods to improve efficiency and quality.



Fig.No.2.2.1 - Challenges and limitations of post-processing techniques in additive manufacturing

2.3 Evaluation Of Manual Vs EDM Support Removal (Ameen et al.. 2022)

NAmeen et al. compared manual support removal with wire Electrical Discharge Machining (EDM) for Ti6Al4V metal parts produced using EBM. Manual removal was faster (average 3.08 minutes) but resulted in poor surface quality ($R_a = 27.25 \mu\text{m}$) and visible residues. Wire EDM provided a smoother surface ($R_a = 4.01 \mu\text{m}$) but required longer processing time (9.88 minutes) and caused a heat-affected zone (HAZ) of $8 \mu\text{m}$. Although EDM offered superior surface finish, it slightly reduced hardness due to thermal effects.

2.4 Emerging Alternatives and Trends

Researchers like Hildreth et al. and Chou et al. developed dissolvable or contact-free support structures, enabling easier removal via chemical or thermal means. Electrochemical etching and unmelted powder bridging techniques allow for contact-free supports that minimize surface damage. These approaches enhance automation and reduce human intervention but are limited to specific materials and higher process costs.

2.5 Research Gap Identified

Existing studies mainly focus on metal additive manufacturing, while polymer-based 3D printing and automated cleaning mechanisms are less explored. There is a need for compact, cost-effective, and automated cleaning systems that can reduce manual intervention while maintaining part integrity.

2.6 Relevance to Current Project

The reviewed studies provide a foundation for developing a Support Cleaning Machine that automates post-processing. Incorporating temperature control, agitation, and safe material handling as inspired by existing research can significantly improve cleaning speed, consistency, and surface quality.

Chapter 3

Problem Statement & Proposed Solution

The problems identified with conventional support removal methods for 3D printed components and solution for the same is mentioned below.

3.1 Problem Statement

In additive manufacturing, support structures are essential for printing complex geometries, overhangs, and intricate designs. However, the removal of these support structures is often performed manually, which creates several challenges.

1. Time Consumption

Manual removal of support structures requires significant time and effort, especially for components having complex geometries. This reduces overall production efficiency and increases post-processing time.

2. Risk Of Component Damage

During manual support removal, delicate features of the printed component may be damaged. Excessive force or improper handling can affect dimensional accuracy and surface quality.

3. Inconsistent Cleaning Performance

Traditional support removal methods depend on operator skill and experience. As a result, the quality of cleaning varies from one component to another, leading to inconsistent results.

4. Safety Issues

Support removal often involves the use of chemical solutions and sharp tools. Direct human interaction with these materials may create safety hazards and increase the possibility of accidents.

5. Lack of Automation

Most low-cost support removal methods lack temperature control, agitation mechanisms, and process monitoring systems, resulting in reduced cleaning efficiency and longer processing time.

3.2 Proposed Solution

To overcome the limitations of conventional support removal methods, an automated support cleaning machine has been developed. The machine utilizes a chemical cleaning solution along with controlled temperature and mechanical agitation to remove soluble support materials effectively.

The system consists of a stainless-steel cleaning tank, electric heater, circulation motor, thermocouple sensor, ESP8266 microcontroller, LCD display, and Blynk-based IoT monitoring system.

The cleaning solution is heated to an optimum temperature range of 50°C to 70°C, which accelerates the dissolution of support material. Simultaneously, the agitation mechanism ensures uniform circulation of the solution around the printed component, resulting in faster and more effective cleaning.

The automated control system continuously monitors the temperature and controls heater operation, ensuring safe and consistent performance. This approach minimizes manual intervention, improves cleaning quality, reduces processing time, and prevents damage to printed parts.

Chapter 4

Objective & Methodology

4.1 Objective

4.1.1 Design and Development of Compact Support Cleaning Machine

The primary objective of this project is to design and develop a compact support cleaning machine for post-processing of 3D printed components. The machine is intended to occupy minimum space while providing effective support removal capabilities. The design includes a cleaning tank, heating system, agitation mechanism, temperature sensing unit, and control system integrated into a single compact structure. The compact configuration makes the machine suitable for educational institutions, research laboratories, prototyping centers, and small-scale industries where space availability is limited.

4.1.2 Develop a Low-Cost and Efficient Cleaning Solution

Another important objective is to develop an economical and efficient support cleaning system that can serve as an alternative to expensive commercial support removal equipment. The project focuses on utilizing readily available components such as ESP8266 microcontroller, electric heater, thermocouple sensor, LCD display, and EcoWorks cleaning solution. The proposed design minimizes manufacturing and operating costs while maintaining high cleaning performance, making it affordable for a wide range of users involved in additive manufacturing.

4.1.3 Dissolve Support Material Using pH-Based Cleaning Solution

The project aims to effectively remove soluble support materials from 3D printed parts using a pH-controlled cleaning solution. The machine utilizes EcoWorks cleaning agent for dissolving support materials such as SR-30 and QSR. Controlled temperature and solution circulation accelerate the dissolution process and ensure uniform cleaning. The objective is to achieve complete support removal while preserving the dimensional accuracy and surface quality of the printed component.

4.1.4 Ensure Safe and Damage-Free Cleaning of Printed Components

A major objective of the project is to provide a safe and reliable cleaning process that minimizes the risk of damage to delicate 3D printed components. Manual support removal often causes

scratches, cracks, and dimensional inaccuracies. The proposed automated system uses controlled heating and agitation mechanisms to remove support structures without applying excessive mechanical force. Temperature monitoring and automatic control further enhance operational safety and process reliability.

4.1.5 Reduce Post-Processing Time and Manual Effort

The project focuses on reducing the time and labor involved in post-processing operations. Traditional support removal methods require considerable manual effort and operator involvement. The developed machine automates the cleaning process through temperature control and continuous agitation, significantly reducing human intervention. This automation improves productivity, shortens processing time, and increases consistency in support removal operations.

4.1.6 IoT-Based Monitoring and Temperature Control

The project also aims to incorporate modern IoT technology for monitoring and controlling machine operation. An ESP8266 NodeMCU microcontroller, thermocouple sensor, LCD display, and Blynk application are integrated into the system to provide real-time temperature monitoring and automatic heater control. This feature improves operational convenience, enhances process accuracy, and enables efficient machine management.

4.1.7 Experimental Testing and Performance Evaluation

The developed support cleaning machine will be experimentally tested using 3D printed components containing soluble support materials. Various operating parameters such as temperature, cleaning duration, and support dissolution efficiency will be evaluated. The collected results will be analyzed to assess system performance and identify opportunities for optimization.

4.1.8 Research Paper Publication and Technical Contribution

The project aims to contribute to the field of additive manufacturing through publication of a research paper. The paper will present the design methodology, component selection, experimental setup, results, and performance analysis of the support cleaning machine. This publication will provide valuable information for researchers and industries involved in 3D printing and post-processing technologies.

4.1.9 Future Development and Commercial Potential

The final objective is to establish a foundation for future improvements and commercialization. The machine can be upgraded with advanced features such as automatic pH monitoring, automatic solution replacement, AI-based process optimization, and remote monitoring systems. These developments can enhance industrial applicability and increase the commercial value of the proposed support cleaning machine.

4.2 Methodology

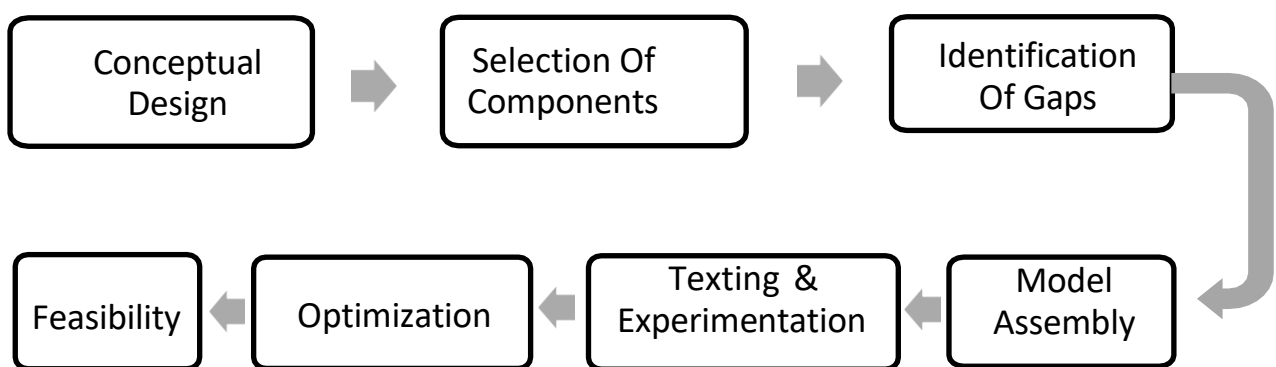


Fig.No.4.2.1 - Methodology of Support Cleaning Machine

1. Conceptual Design

The initial design concept was developed based on project requirements, functionality, and user needs.

2. Selection of Components

Suitable components and materials were selected considering performance, availability, cost, and compatibility.

3. Identification of Gaps

Potential design limitations, operational challenges, and improvement areas were identified through analysis.

4. Model Assembly

All selected components were assembled to create the prototype/model according to the design specifications.

5. Testing & Experimentation

The assembled model was tested under different operating conditions to evaluate its performance and reliability.

6. Optimization

Based on test results, necessary modifications were made to improve efficiency, functionality, and overall performance.

7. Feasibility Analysis

The final design was evaluated for technical, economic, and practical feasibility to ensure successful implementation.

Chapter 5

Conceptual Design & Modelling

5.1 Conceptual Design

The below conceptual design is made with a SolidWorks software.

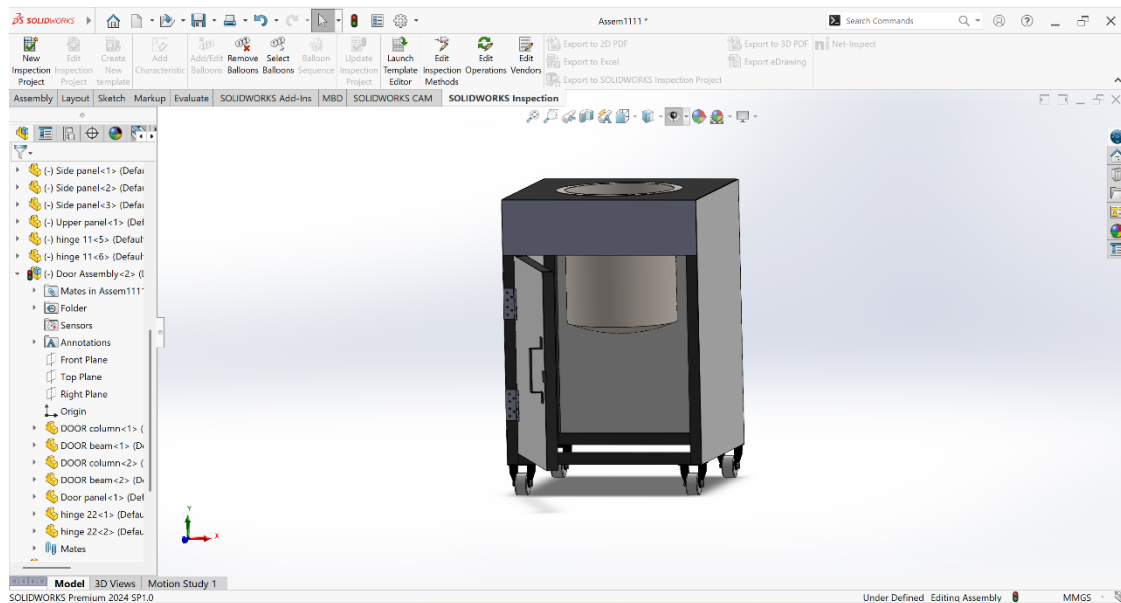


Fig. No.5.1.1 - Conceptual Design of support cleaning machine

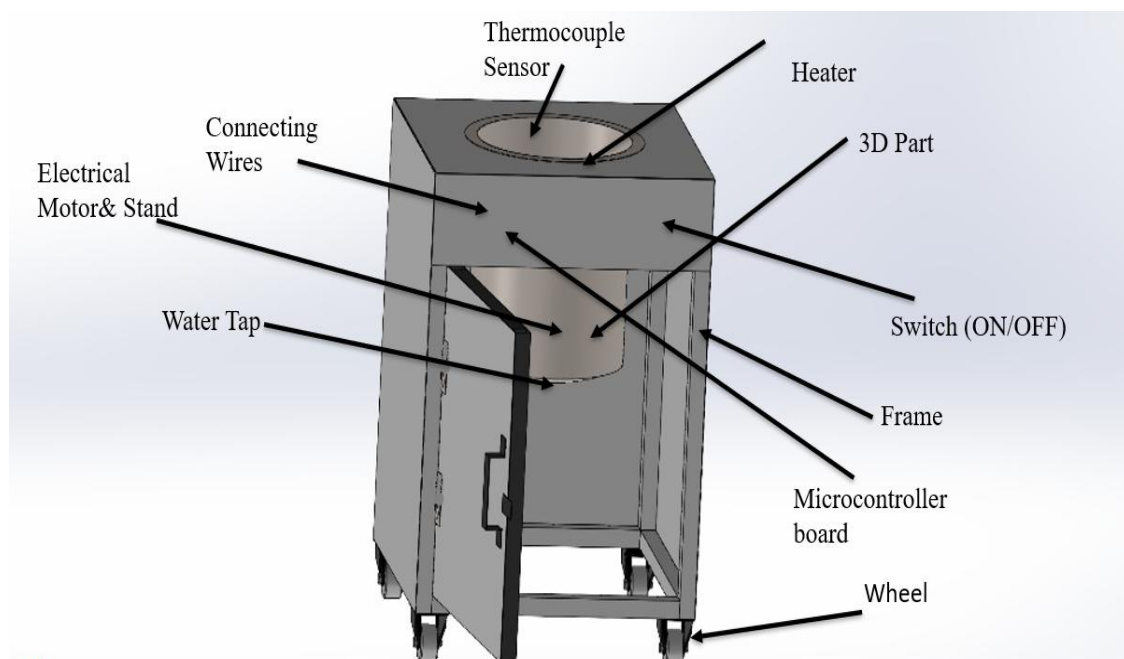


Fig. No.5.1.2 - Conceptual Design of support cleaning machine 3d Printer

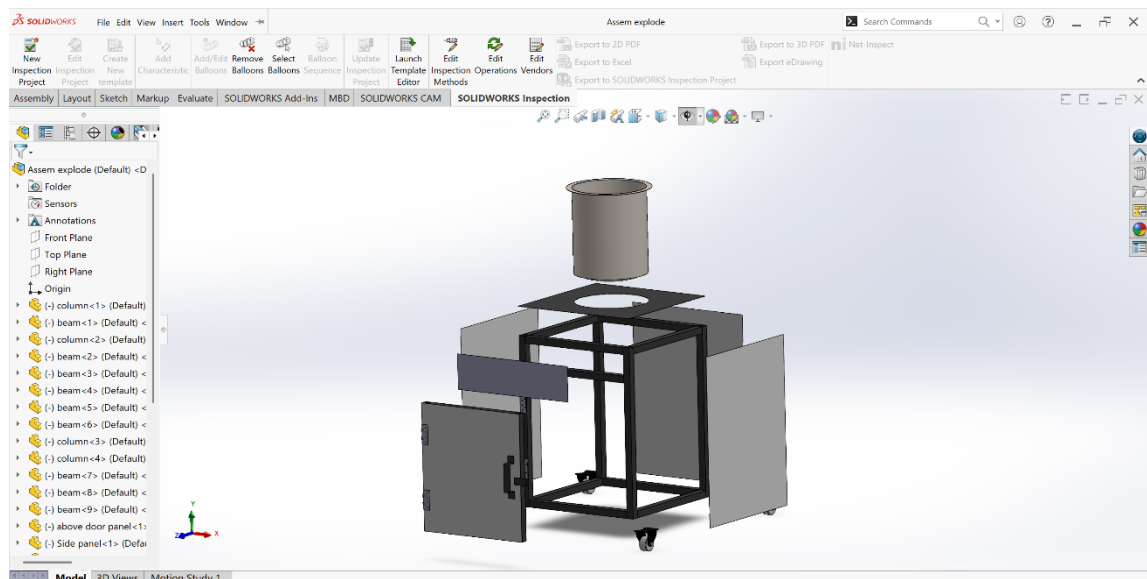


Fig. No.5.1.3 - Conceptual Design Exploded View

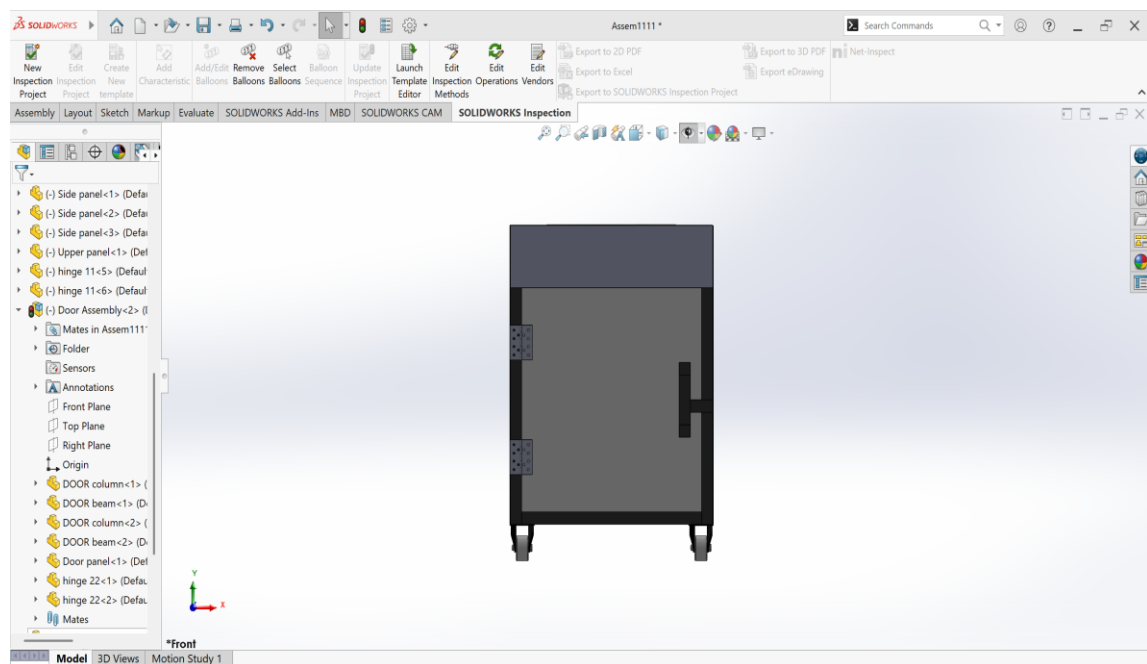


Fig. No.5.1.4 - Conceptual Design Front View

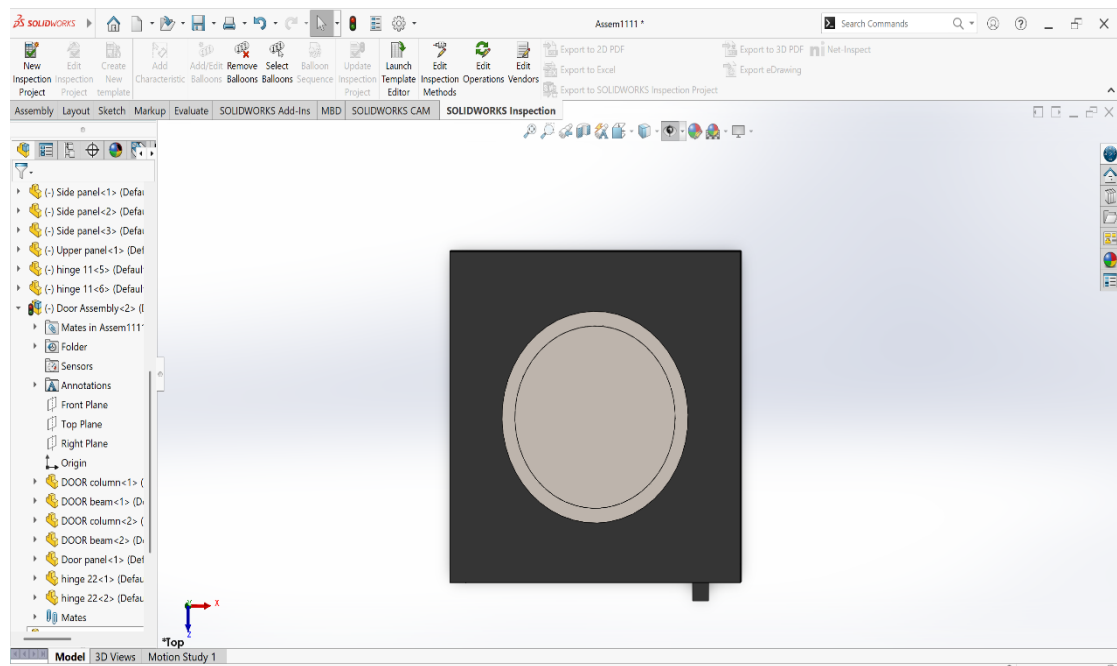


Fig. No.5.1.5 - Conceptual Design Top View

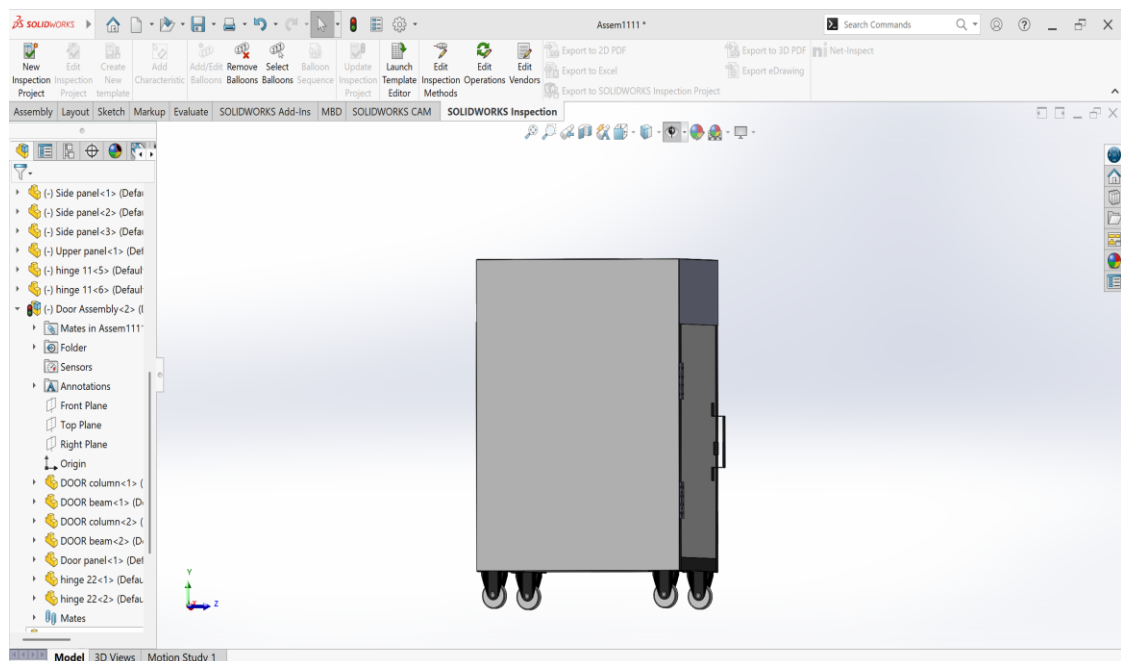


Fig. No.5.1.6 - Conceptual Design Right Side View

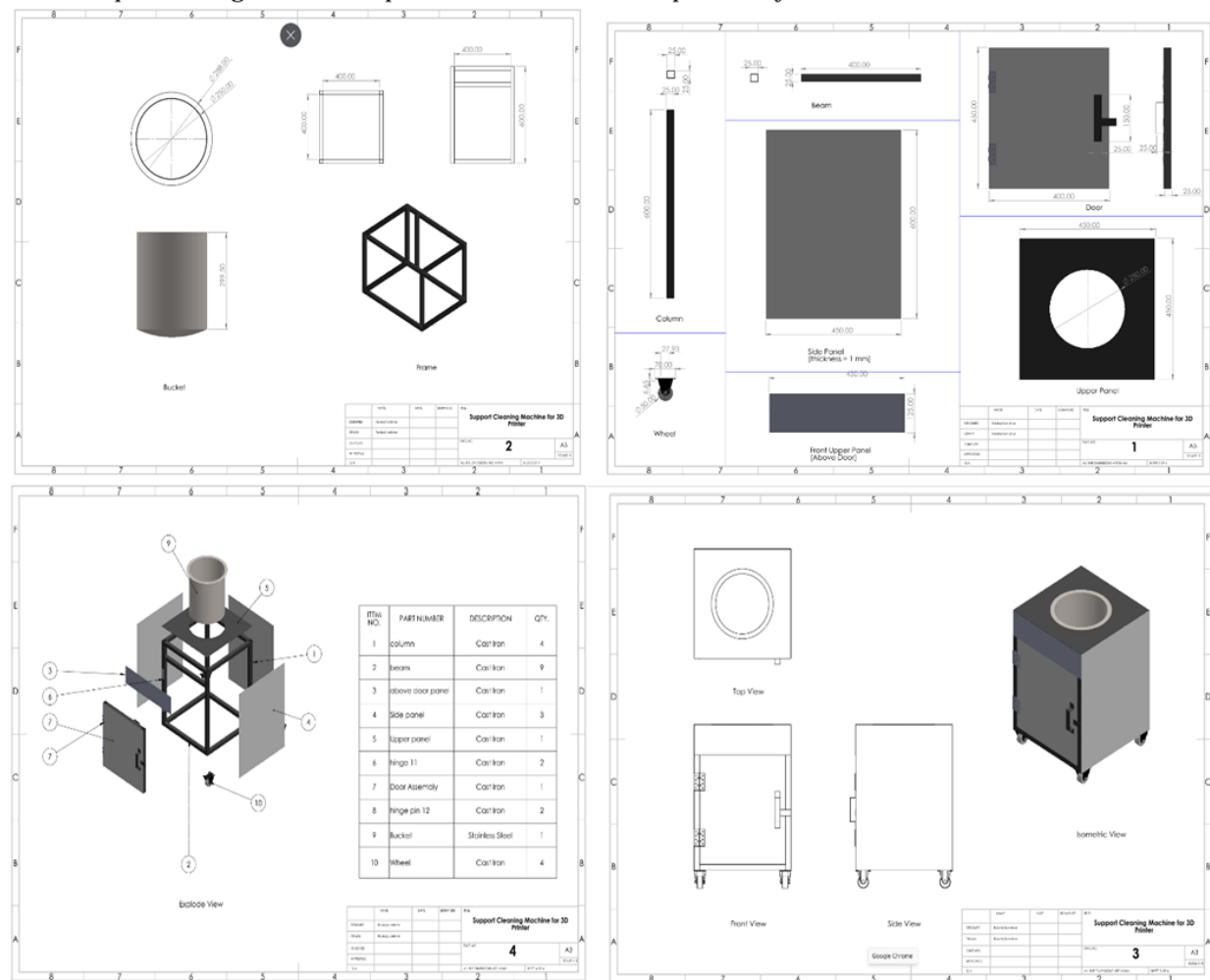


Fig. No.5.1.7 – Drawings of support cleaning machine

5.2 Components Selection

In this stage, the appropriate mechanical and electrical components are carefully chosen based on the design requirements of the support cleaning machine. Material selection for the tank, frame, and other structural parts is carried out considering strength, durability, and corrosion resistance. A stainless-steel tank is selected to ensure chemical compatibility with the cleaning solvent, while a mild steel frame is used for providing robust structural support. Heating elements are chosen to maintain the solvent temperature between 50°C and 70°C, which is the optimal range for dissolving support materials. To ensure precise control, temperature sensors and an Arduino ESP8266 module are integrated for automation and monitoring. A DC motor or stirrer is incorporated to create uniform agitation within the tank, improving cleaning efficiency. Components are selected based on availability, cost-effectiveness, and performance reliability, while safety elements such as fuses, relays, and proper insulation are included to prevent electrical or thermal hazards. The cleaning agent SR-30 (EcoWorks) is selected for its excellent ability to dissolve soluble supports without damaging the printed part surface. Each component's power rating, dimensions, and compatibility are analyzed, and CAD models are updated with precise specifications. The electronic control system is designed for automatic operation, enhancing user convenience.

1. Frame

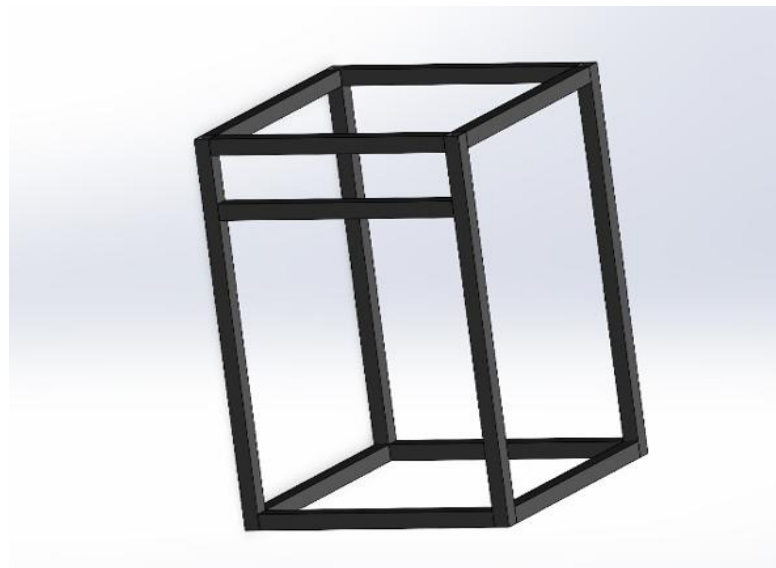


Fig. No.5.2.1 – Frame

The frame serves as the main supporting structure of the Support Cleaning Machine for 3D Printers. It is made from square hollow steel sections to provide sufficient strength and stability. The frame is designed to securely hold all machine components in their proper positions while reducing vibrations

during operation. Its simple and sturdy construction makes fabrication, assembly, and maintenance easy. The compact design also helps in efficient use of space and ensures reliable machine performance.

2. Heater Coil



Fig. No.5.2.2 – Heater Coil

The heating element is used to generate the required heat for the support removal process. It converts electrical energy into thermal energy and helps soften or weaken the support material attached to the 3D printed part. This makes the cleaning and removal process easier, faster, and more efficient. The heating element is designed to provide uniform heat distribution while ensuring safe and reliable operation of the machine.

3. Water Tank



Fig. No.5.2.3 – Stainless Steel Water Tank

The stainless steel water tank is used to hold the water required for the support removal process. It acts as the main container where the 3D printed parts are immersed during cleaning. Stainless steel is selected due to its excellent corrosion resistance, durability, and ability to withstand elevated temperatures. The tank ensures safe containment of water and provides a reliable working environment for efficient support removal while maintaining cleanliness and long service life.

4. LCD Display



Fig. No.5.2.4 – LCD Display

In the 3D Printer Support Removal Machine, the 16x2 I2C LCD Display is used to provide real-time information about the cleaning process. It displays parameters such as operating status, temperature, and system notifications. The display helps the operator monitor the machine's performance and ensures smooth operation. The I2C interface reduces wiring complexity and allows easy integration with the microcontroller, making the system more compact and efficient.

5. Microcontroller Board (NodeMCU ESP8266)



Fig. No.5.2.5 – Microcontroller Board (NodeMCU ESP8266)

The microcontroller board (NodeMCU ESP8266) acts as the central control unit of the Support Cleaning Machine. It is responsible for monitoring and controlling various machine functions such as temperature regulation, heating element operation, and overall system coordination. The board processes input signals from sensors and executes the programmed logic to ensure efficient and safe operation. Its compact size, low power consumption, and built-in Wi-Fi capability make it suitable for automation and future smart monitoring applications.

The ESP8266 NodeMCU is a Wi-Fi-enabled microcontroller board applications, featuring built-in Wi-Fi and easy programming via USB (CP2102). It allows wireless data transmission, sensor monitoring, and automation.

6. Thermocouple Sensor

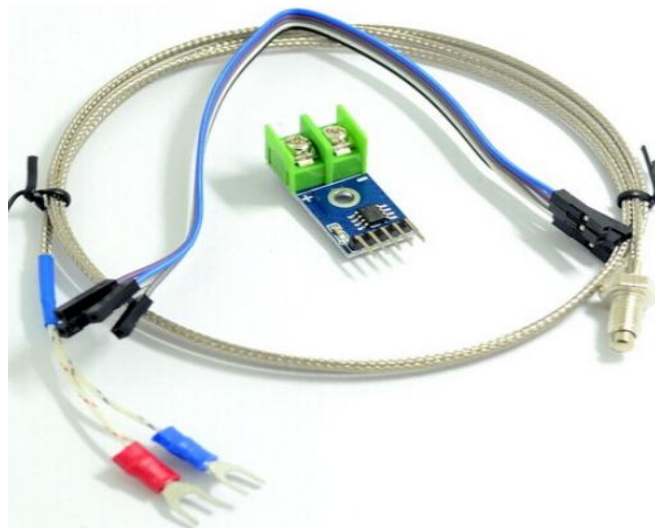


Fig. No.5.2.6 – Thermocouple Sensor

The thermocouple sensor is used to measure and monitor the temperature of the water inside the tank. It provides real-time temperature data to the microcontroller, enabling accurate control of the heating element. This helps maintain the desired temperature required for effective support removal from 3D printed parts. The sensor offers fast response, reliable performance, and high-temperature measurement capability, ensuring safe and efficient operation of the system.

7. Stirrer Motor



Fig. No.5.2.7 – Stirrer Motor

The stirrer motor is used to rotate the stirring mechanism inside the water tank. It creates continuous movement of water, ensuring uniform temperature distribution and improving the effectiveness of the support removal process. The motor helps increase the contact between the heated water and the 3D printed parts, resulting in faster and more efficient cleaning. Its compact design, reliable operation, and low power consumption make it suitable for continuous use in the machine.

8. ON – OFF Switch



Fig. No.5.2.8 – ON – OFF Switch

The ON–OFF switch is used to control the power supply of the Support Cleaning Machine. It allows the user to safely start or stop the operation of the system whenever required.

Chapter 6

EXPERIMENTAL AND CFD APPROACH

The Experimental and CFD approach was adopted to analyze, design, and validate the performance of the Support Cleaning Machine for 3D printers. The experimental setup was developed to study the effect of temperature, agitation, and cleaning time on the dissolution rate of support materials such as QSR-30. The fabricated prototype consisted of a stainless steel cleaning tank, heating element, temperature control system, and agitation mechanism. The system was tested under different temperature ranges from 50°C to 70°C, ensuring uniform heating and efficient cleaning performance. To ensure thermal stability and uniform heat distribution, Computational Fluid Dynamics (CFD) simulations were conducted using the 3D CAD model of the tank.

6.1 Introduction

The experimental and computational analysis (CFD) of the support cleaning machine is essential to evaluate its performance, validate the design parameters, and optimize operational conditions. The experimental approach focuses on prototype fabrication, component integration, and performance testing, while the CFD analysis provides insight into fluid flow, temperature distribution, and heat transfer characteristics inside the cleaning tank. This dual methodology enables a comprehensive understanding of how the system behaves under actual and simulated conditions. Experimentally, the machine's operation is observed by measure in temperature profiles, clean in time, and power consumption. The results are used to verify whether the cleaning solution reaches uniform temperature and if agitation effectively enhances dissolution. On the computational side, the CFD model replicates the internal fluid dynamics using ANSYS Fluent or SolidWorks Flow Simulation, considering parameters such as tank geometry, heater placement, and flow circulation. Boundary conditions like inlet velocity, fluid properties, and heat flux are defined based on experimental data to ensure realistic simulation. The CFD results help identify zones of poor circulation or heat loss, enabling design improvements such as optimized heater location or modified agitation direction.

6.2 Components of the Machine

The design and fabrication process of the support cleaning machine involve translating conceptual CAD models into a fully functional prototype. The design was carried out using SolidWorks, with emphasis on structural stability, ease of assembly, and compactness. The machine consists of four

major subsystems — the cleaning tank, heater assembly, frame, and control unit — integrated into a single operational platform. The frame and tank dimensions were optimized to accommodate medium sized 3D-printed parts while ensuring ergonomic operation. Fabrication began with the mild steel frame construction, followed by mounting the stainless steel cleaning tank (SS304) to resist corrosion. The heating element was installed beneath the tank with proper insulation to minimize heat loss. Electrical components, including the temperature sensor, PID controller, and power circuit, were wired inside a protected control box. The entire structure was powder-coated to prevent rusting and give a laboratory-grade finish. During assembly, special attention was given to vibration isolation and leak-proof sealing. Gaskets and PTFE tape were used at all joints to prevent fluid leakage. The fabricated unit underwent preliminary dry testing to check electrical safety, followed by wet testing with actual cleaning agents. The prototype demonstrated stable heating performance and consistent temperature control. This design-fabrication process established a solid foundation for further experimental analysis and performance optimization.

6.3 Design and Fabrication of Support Cleaning Machine

The support cleaning machine is composed of multiple integrated components, each serving a specific function in achieving efficient support removal. The major components include the cleaning tank, machine frame, temperature sensor and control circuit, and cleaning solution system. Each component was selected based on its mechanical strength, chemical compatibility, and operational reliability. The cleaning tank houses the support material and cleaning solution, where the main reaction occurs. The frame provides structural rigidity and stability to withstand vibration during agitation. The temperature sensor and control circuit maintain the desired thermal conditions, while the cleaning solution ensures safe and efficient dissolution of the support material. Other auxiliary components include a drain valve, agitation mechanism, and power supply system for reliable operation. Each part was carefully positioned in the CAD model to ensure space optimization and ease of maintenance. This integration of mechanical, electrical, and chemical subsystems allows the machine to operate seamlessly, achieving consistent cleaning performance with minimal human intervention.

6.4 Instrumentation

The Instrumentation involves the measurement, monitoring, and control systems integrated into the machine for performance evaluation and operational safety. The primary instruments include temperature sensors, ammeters, voltmeters, relays, digital controllers, and timers. These

components ensure precise control over process variables such as temperature, current, and voltage during cleaning operations. Instrumentation wiring follows a structured layout, segregating high-voltage and low-voltage lines to minimize interference. The control system records real-time temperature fluctuations, enabling verification of heating uniformity and energy efficiency. Calibration of sensors is performed to ensure accurate data acquisition during testing. For data logging and analysis, digital instruments or microcontroller-based systems can interface with computerized monitoring software. Safety instruments such as fuse blocks, earthing points, and insulation breakers are incorporated to comply with laboratory safety standards.

6.4.1 Temperature Measurement and Control Calibration

Temperature calibration ensures that the readings from the sensor and control system accurately reflect the actual solution temperature inside the tank. Calibration is carried out using a reference thermometer or a digital temperature calibrator. The procedure involves gradually heating the solution while comparing the sensor output with the reference value at multiple temperature points. Any detected offset is corrected through the PID controller settings or firmware adjustments. The calibration is repeated several times to confirm sensor stability and repeatability. Proper calibration minimizes error in temperature control, which directly affects the cleaning efficiency and energy usage. The calibration setup also includes data logging to analyze the heating curve, response time, and overshoot behavior of the system. A well-calibrated control ensures uniform temperature throughout the tank, preventing overheating and guaranteeing consistent cleaning results. Regular recalibration after several operational cycles maintains long-term accuracy and system reliability.

6.4.2 Power Supply and Electrical Safety Setup

The power supply system provides safe and stable electrical power to operate the heater, control circuit, and other components. The machine works on a 230V AC single-phase supply with proper earthing and protection. MCBs, fuses, and relays are used to prevent overloads and short circuits. An SSR controls the heater efficiently, while insulated wiring and emergency stop features ensure operator safety. Electrical components are enclosed in protective casings with ventilation, and digital meters are provided for monitoring voltage and current.

Chapter 7

Prototype & Assembly

7.1 Assembly

The assembly of the Support Cleaning Machine involves the systematic integration of mechanical, electrical, and electronic components into a compact and efficient cleaning system. The process begins with the fabrication and assembly of the mild steel frame, which serves as the primary supporting structure of the machine. A stainless steel (SS304) cleaning tank is securely mounted on the frame to provide corrosion resistance and durability during operation.

The heating element is installed beneath the cleaning tank to ensure uniform heat transfer to the cleaning solution. A circulation motor is mounted inside the tank assembly to provide continuous agitation of the cleaning solution, improving the dissolution rate of support materials. The thermocouple sensor is positioned appropriately to continuously monitor the solution temperature and provide accurate feedback to the control system.

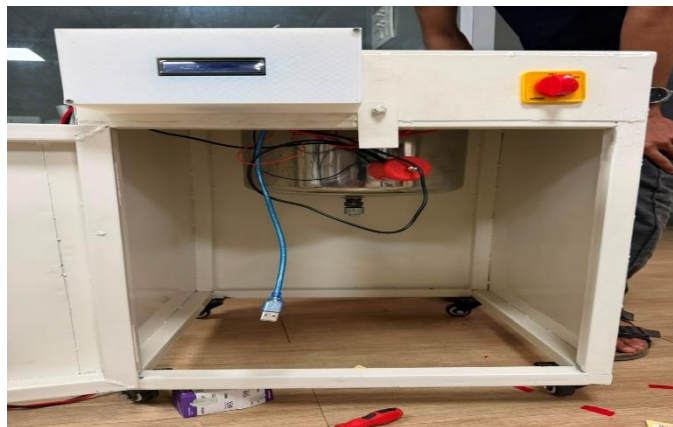


Fig. No.7.1.1 – Assembly

Electronic components such as the ESP8266 NodeMCU controller, LCD display, relay module, switches, and power supply unit are mounted in accessible locations. The ESP8266 acts as the main controller, receiving temperature data from the thermocouple sensor and controlling the heater through relay circuits. The LCD display provides real-time temperature information, while the Blynk platform enables remote monitoring.

All wiring and cable connections are arranged properly to ensure safety and ease of maintenance. Each component is individually tested before final assembly to verify proper operation. The

completed assembly results in a compact, stable, and efficient support cleaning machine capable of automated support removal from 3D printed components.



Fig. No.7.1.2 – Assembly Front View



Fig. No.7.1.3 – Assembly Top View



Fig. No.7.1.4 – Assembly Side View

7.2 Commands & Coding

```
#define BLYNK_TEMPLATE_ID "TMPL3cfMYDy_d"
#define BLYNK_TEMPLATE_NAME "Support Cleaning Machine"
#define BLYNK_AUTH_TOKEN "Alc9o58glo8QJ0w8WdGDXH1LJ9Cvs0E6"
```

```
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include "max6675.h"
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "iot";
char pass[] = "123456789";
int thermoDO = 12;
int thermoCS = 15;
int thermoCLK = 14;
#define RELAY1 D3 // Mumbai
#define RELAY2 D4 // Pune
```

```
MAX6675 thermocouple(thermoCLK, thermoCS, thermoDO);
int a;
BLYNK_WRITE(V1){
  a = param.asInt();
  if (a==1){
    digitalWrite(RELAY1,0);
  }
  else{
    digitalWrite(RELAY1,1);
  }
}
```

```
}  
}  
  
void setup() {  
  Wire.begin(D2, D1);  
  lcd.begin(16,2);  
  lcd.backlight();  
  
  Serial.begin(115200);  
  Blynk.begin(auth,ssid,pass);  
  pinMode(RELAY1,OUTPUT);  
  pinMode(RELAY2,OUTPUT);  
  digitalWrite(RELAY1,1);  
  digitalWrite(RELAY2,1);  
  lcd.setCursor(0, 0);  
  lcd.print("WELCOME TO");  
  lcd.setCursor(0, 1);  
  lcd.print("VPKBIET");  
  delay(5000);  
  lcd.clear();  
  delay(1000);  
  lcd.setCursor(0, 0);  
  lcd.print("This is Support");  
  lcd.setCursor(0, 1);  
  lcd.print("Cleaning Machine");  
  delay(5000);  
  lcd.clear();  
  delay(1000);  
  
  Serial.println("MAX6675 test");  
  // wait for MAX chip to stabilize  
  delay(500);  
}
```

```
void loop() {  
  // basic readout test, just print the current temp  
  
  Serial.print("C = ");  
  Serial.println(thermocouple.readCelsius());  
  Blynk.virtualWrite(V0,thermocouple.readCelsius());  
  lcd.setCursor(0, 0);  
  lcd.print("Temp:");  
  lcd.setCursor(6, 1);  
  lcd.print(thermocouple.readCelsius());  
  if(thermocouple.readCelsius()<=70){  
    digitalWrite(RELAY2,0);  
  }else{  
    digitalWrite(RELAY2,1);  
  }  
  delay(1000);  
}
```

7.3 Electronic Integration

The electronic system of the Support Cleaning Machine is centered around the ESP8266 NodeMCU microcontroller. The thermocouple sensor continuously measures the cleaning solution temperature and transmits data to the controller through the MAX6675 interface module. The ESP8266 processes the temperature data and controls the heater using relay modules. The LCD display is integrated with the controller to provide real-time temperature information to the operator. A circulation motor is also connected to ensure continuous agitation of the cleaning solution.

The Blynk IoT platform communicates with the ESP8266 through Wi-Fi, enabling remote monitoring and system control. The integration of sensors, relays, display units, and communication modules creates an automated and efficient support cleaning system.



Fig. No.7.3.1 – LCD Fitting



Fig. No.7.3.2 – Thermocouple Wiring



Fig. No.7.3.3 – Stirrer & Stand Fitting



Fig. No.7.3.4 – On – Off Knob Connection



Fig. No.7.3.5 – ESP8266 & LCD integration

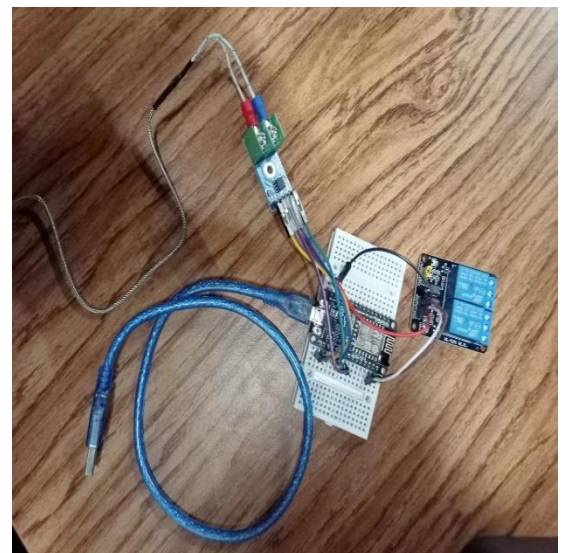


Fig. No.7.3.6 – ESP8266 & Microcontroller

7.4 User Interface & Monitoring

The Support Cleaning Machine is equipped with an LCD display and IoT-based monitoring system. The LCD display The Blynk application acts as a user-friendly monitoring platform, allowing users to observe operating conditions remotely through a smartphone or computer. The interface displays temperature readings, heater status, and machine operation details in real time.



Fig. No.7.4.1 – User Interface



Fig. No.7.4.2 – User Interface



Fig. No.7.4.3 – User Interface (Temperature 42°C)

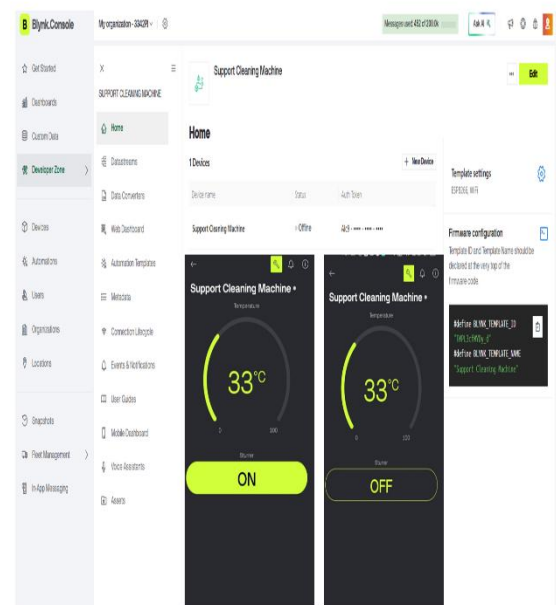


Fig. No.7.4.4 – Blink App Interface

7.5 Circuit Diagram

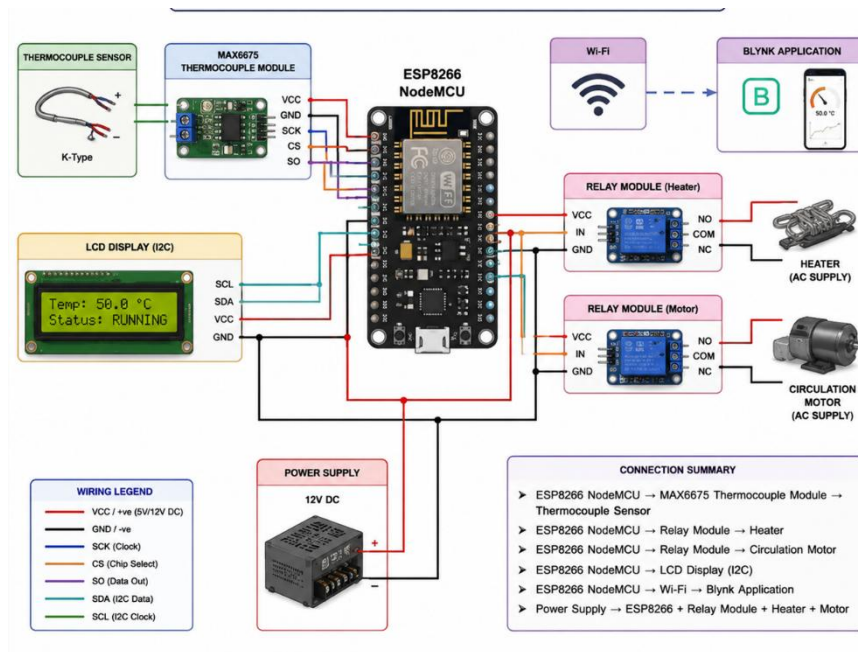


Fig. No.7.5.1 – Circuit Diagram

Chapter 8

Result & Outcome

8.1 Result

8.1.1 Support Material Removal Efficiency

The developed support cleaning machine successfully removed soluble support materials such as SR-30 and QSR from 3D printed components. The cleaning process was effective and provided consistent results across multiple trials.

8.1.2 Temperature Control Performance

The heating system maintained the cleaning solution temperature within the desired range of 50°C to 70°C. The thermocouple sensor and controller ensured accurate temperature monitoring and stable operation.

8.1.3 Cleaning Time Reduction

The automated cleaning process significantly reduced the time required for support removal compared to conventional manual methods. Continuous agitation and controlled temperature accelerated support dissolution.

8.1.4 Safe and Damage-Free Cleaning

The machine removed support structures without causing damage to the printed components. Surface quality and dimensional accuracy of the parts were maintained throughout the cleaning process.

8.1.5 IoT Monitoring Performance

The ESP8266 microcontroller and Blynk platform successfully monitored temperature data in real time. The system provided reliable operation and improved process control.

8.2 Outcome

8.2.1 Working Prototype Development

A fully functional prototype of the Support Cleaning Machine was successfully designed,

fabricated, and tested for post-processing of 3D printed components.

8.2.2 Automated Support Removal System

The project demonstrated an automated method for removing soluble support materials, reducing manual effort and improving cleaning consistency.

8.2.3 Integration of Mechanical and Electronic Systems

The project successfully integrated the cleaning tank, heater, motor, thermocouple sensor, ESP8266 controller, LCD display, and Blynk platform into a single system.

8.2.4 Enhanced Understanding of Additive Manufacturing Post-Processing

The project provided practical knowledge of support removal techniques, temperature control systems, IoT applications, and post-processing requirements in additive manufacturing.

8.2.5 Potential Industrial Applications Identified

The developed machine can be utilized in educational institutions, research laboratories, prototyping centers, and small-scale industries involved in 3D printing and additive manufacturing.



Fig. No.8.1 – Component with Supports
(Before Cleaning)



Fig. No.8.2 – Component without Supports
(After Cleaning)

Chapter 9

Timeline & Expenditure

9.1 Timeline

ACTIVITY	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR, APRIL ,MAY, JUNE
Literature Review on 3D Printing and Support Removal Techniques									
Study of Existing Support Cleaning Machines and Technologies									
Identification of Design Requirements and Components									
Conceptual Design and CAD Modelling in SolidWorks									
Design Verification and Model Optimization									
Procurement of Mechanical, Electrical and Electronic Components									
Fabrication of Frame, Tank and Supporting Structure									
Assembly of Mechanical, Electrical and Electronic Components									
Programming of ESP8266, Sensor Integration and Blynk Setup									
Experimental Testing, Performance Evaluation and Optimization									
Research Paper Publication and Conference Presentation									
Project Report Writing, Documentation, Project Compilation and Final Submission									

Table 9.1 - Timeline

9.2 Expenditure

Sr. No.	Components	Quantity	Approx. Amount (Rs.)
1	Frame	1	3000/-
2	Heater	1	550/-
3	Water Tank	1	500/-
4	LCD Display	1	170/-
5	Microcontroller Board	1	260/-
6	Thermocouple Sensor	1	340/-
7	Electrical Cooling Motor	1	150/-
8	Connecting Wire,And Cables	-	740/-
9	Switch Button	1	150/-
10	Stand	1	180/-
11	Water Tap & Gas Welding	1	260/-
Total			6300/-

Table 9.2 - Expenditure Sheet

Chapter 10

Conclusion

This project “Design and Development of Support Cleaning Machine for 3D Printer” successfully achieved its objective of automating the removal of support materials from 3D-printed components. The developed system effectively cleans medium-sized printed parts using EcoWorks cleaning solution and soluble support materials such as SR-30. The SS304 cleaning tank provides excellent corrosion resistance and uniform heat transfer, while the mild steel frame ensures structural strength and stability. The integrated heating system maintains an optimum temperature range of 50°C–70°C for efficient support dissolution. Temperature monitoring and control improve process consistency and reliability. Experimental results showed improved cleaning efficiency and reduced processing time. The machine minimizes manual effort and reduces operator exposure to cleaning chemicals. Its low-cost and compact design makes it suitable for educational institutions, laboratories, and small-scale industries. The project provides a reliable foundation for future advancements such as IoT-based monitoring and automated process control.



Fig. No.10.1 – Photo With Project Guide

Chapter 11

Future Scope

Advanced Automation and Intelligent Process Control

One significant area for future development of the Support Cleaning Machine is the implementation of advanced automation and intelligent process control systems. Currently, the machine operates using temperature monitoring and manual preparation of the cleaning solution. Future versions can incorporate automatic pH monitoring and dosing systems to continuously maintain the optimum chemical concentration required for support material dissolution. This would improve cleaning efficiency and reduce operator involvement.

The integration of IoT technology can further enhance the machine by enabling remote monitoring and control through mobile applications or cloud-based platforms. Users would be able to monitor temperature, cleaning duration, solution condition, and machine status in real time from any location. Such capabilities would improve operational convenience and facilitate predictive maintenance. Another promising advancement is the incorporation of ultrasonic cleaning technology. Ultrasonic vibrations can significantly improve the dissolution rate of support materials by enhancing the interaction between the cleaning solution and the printed component. This would reduce cleaning time and improve the quality of post-processing, especially for parts with complex geometries and internal cavities.

Artificial Intelligence (AI) and machine learning algorithms can also be integrated to automatically optimize cleaning parameters based on material type, support structure density, and component geometry. The system could intelligently adjust temperature, cleaning duration, and agitation speed to achieve the best possible cleaning performance while minimizing energy consumption.

Additionally, a fully automated fluid replacement and self-cleaning mechanism can be developed to reduce maintenance requirements and improve operator safety. Such features would make the machine highly suitable for industrial-scale additive manufacturing applications. Overall, these future enhancements would transform the support cleaning machine into a smart, efficient, and fully automated post-processing system, contributing to the advancement of Industry 4.0 and modern additive manufacturing technologies.

Chapter 12

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Chapter 13

Research Paper

13.1 Publication Details

- **Research Paper Title:** Design And Development Of Support Cleaning Machine For 3D Printer.
- **Author:** Ajinkya S. Chavan, Raviraj V. Kumbhar, Swapnil B. Yadav,
- **Conference:** International Conference on Recent Trends in Technology & Computing (ICRTSTC 2026) Held on 13th and 14th March 2026
- **Organized by:** P Vasundhara Bahuddeshiya Samajik Sanstha's Siddhivinayak Technical Campus Engineering.
- **Paper Details:** The research paper presents the design and development of a low-cost automated Support Cleaning Machine for 3D printed components. The proposed system utilizes an SS304 cleaning tank, controlled heating mechanism, mechanical agitation, and an ESP8266-based temperature monitoring system integrated with the Blynk IoT platform. The machine effectively removes soluble support materials such as SR-30 and QSR using an EcoWorks cleaning solution while maintaining an operating temperature of 50°C–70°C. The developed system minimizes manual effort, reduces cleaning time, improves cleaning consistency, and prevents damage to delicate 3D printed parts. The research highlights a compact, economical, and reliable post-processing solution suitable for educational institutions, research laboratories, prototyping centers, and small-scale additive manufacturing industries. The work also demonstrates the potential for future enhancements through IoT integration, intelligent process control, and advanced automated post-processing technologies.
- **ISBN:** 2348-0831
- **Paper ID:** STC-26-ME011
- **Conference Signatories:** Prof. P.S. Deshmukh (IQAC Coordinator, ICRTSTC 2026)

Dr. A. G. Kulkarni – Principal .

13.2 Paper

International Organization of Research & Development (IORD)
ISSN: 2348-0831
Vol 13 Special Issue 02 | 2026



Design And Development Of Support Cleaning Machine For 3D Printer

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DOI: 10.5281/zenodo.19661824

ABSTRACT

The project "Design and Development of Support Cleaning Machine for 3D Printer" aims to create an efficient, low-cost post-processing system for removing support structures from 3D-printed parts. In additive manufacturing, supports are essential but their manual removal is tedious, inconsistent, and may damage delicate components. The proposed machine automates this process using a chemical-based cleaning system integrated with temperature control and agitation for effective support removal. The design procedure included concept generation, component selection, CAD modeling in SolidWorks, gap identification, and feasibility analysis. The optimized system ensures safe, uniform, and damage-free cleaning. Overall, the machine provides a compact, user-friendly, and sustainable solution that enhances post-processing efficiency and supports the advancement of additive manufacturing technology.

Keywords: - 3D Printing, Support Removal, Support Cleaning Machine, Chemical Cleaning System, Automated Cleaning, Temperature Control, Agitation Mechanism, CAD Design, SolidWorks, Cost-Effective Design, Sustainable Manufacturing.

1. INTRODUCTION

The Additive Manufacturing, or 3D printing, process offers the capability to create complex and intricate parts, but it also results in support structures that need to be removed later. Overall, this process - the manual removal of these support structures - is long, inconsistent, and can be detrimental to sensitive parts. In an effort to eliminate all these drawbacks associated with the post-processing of parts made by the Additive Manufacturing process, a Support Cleaning Machine has been designed that uses a combination of the chemical cleaning medium (QSR-30), temperatures, and mechanical agitation. The machine efficiently dissolves soluble supports without sacrificing part quality, such as PVA and QSR-30. Compact, yet still affordable and user-friendly, with reduced manual effort and improvement in surface finish, this innovative machine integrates mechanical, electrical, and thermal systems to enhance efficiency and reliability in 3D printing workflows.

2. LITERATURE REVIEW

A completely automated technique for the removal of supports was proposed by Nelaturi et al. [1] as part of multi-axis machining systems. The technique is based on the application of recursive algorithms that provide the optimum sequence of removal of the support structure by computing the accessibility of the tools without the occurrence of any collision.

In the comparison of the support removal process via the manual method and Wire Electrical Discharge Machining, Ameen et al. [2] used EBM to produce Ti6Al4V metal components and found the manual method to be more efficient in terms of processing time, averaging 3.08 minutes of processing in contrast to Wire EDM, which took more time (9.88 minutes).

Saigopal Nelaturi.[3] Automatic Support Removal for Additive Manufacturing Post Processing is a research study that focuses on minimizing the time taken for the support structure removal process from the additively manufactured product. The research study presents a solution for the problem using automation, which relies on computer tools to remove the support structure from the product without any damage to the product geometry.

Within the context of this research, the existing methods for support material removal in the post-processing stage in additive manufacturing have been reviewed. As per the existing research, it has been proven that the existing tools for post-processing in the additive manufacturing process have the potential to reduce the human involvement required in the post-processing stage with the help of the existing machining tools, which have the potential for accuracy and precision in the post-processing stage for the finished products using the above overall machining process, even though the processing time in the post-processing stage for the finished products is very less for the employees in the existing machining tools used for the post-processing stage in the machining process.

13.3 Certificates



Fig. No.13.3.1 – Certificate of Ajinkya Chavan



Fig.No.13.3.2 – Certificate of Raviraj Kumbhar



Fig. No.13.3.3 – Certificate of Swapnil Yadav

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A lowcost automated support cleaning machine for 3D prints using chemical cleaning, temperature control, and agitation to ensure safe, uniform, and damagefree postprocessing in additive manufacturing.							
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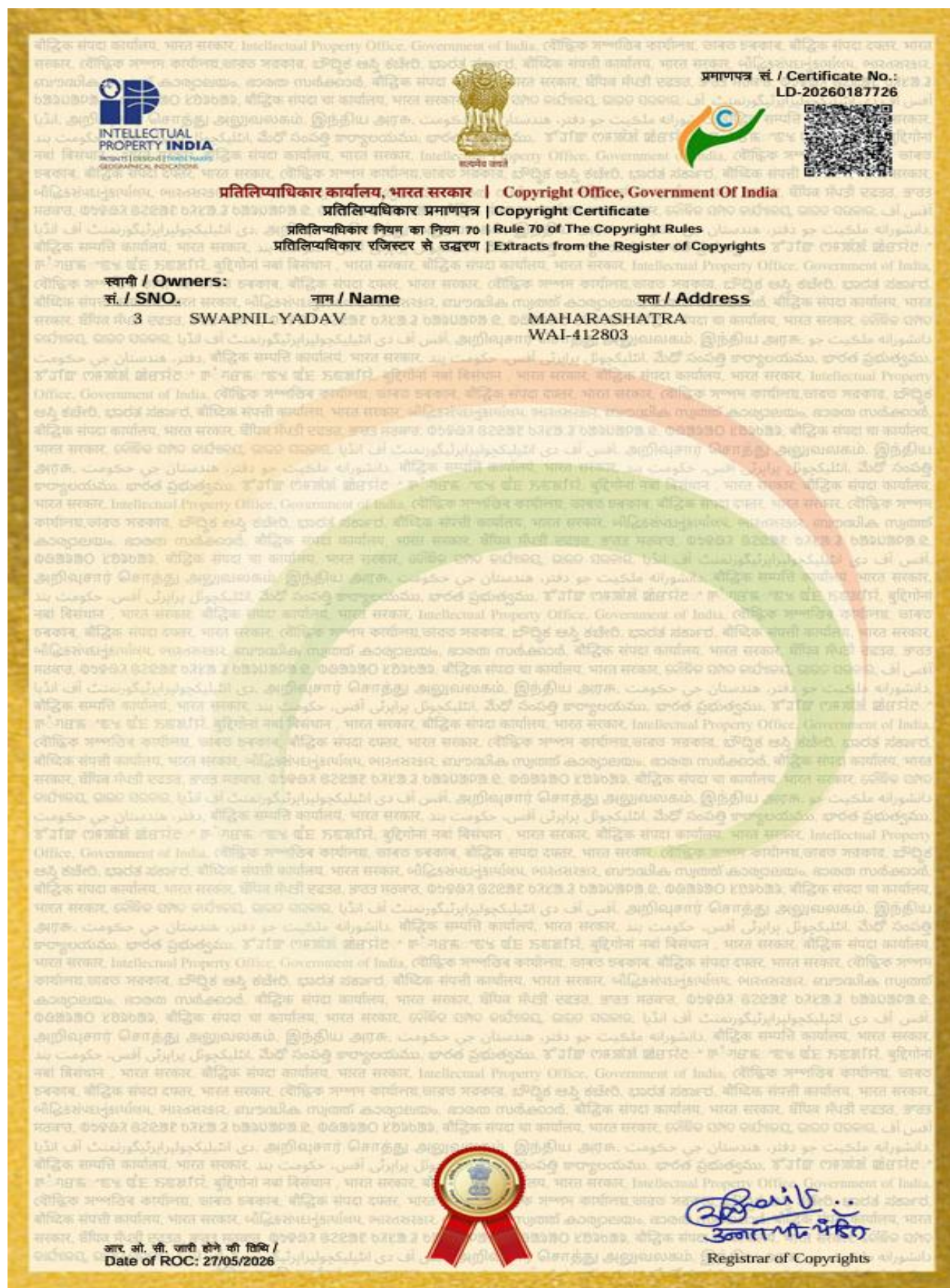


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